Appendix: GRoFA — Noise-Gated Adapters for Jointly Fair and Robust Face Embeddings

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A Cross-Architecture Backbone Results

This appendix provides comprehensive evaluation results for CLIP ViT-B/16 and DINOv2-base backbones, trained with the same GRoFA adapter as the primary BLIP configuration. Section A.1 presents severity-averaged comparison tables with all evaluated methods. Section A.2 reports the WiSE-FT α sweep. Section A.3 compares fairness metrics.

A.1 Severity-Averaged Comparison with All Methods

Tables 1–4 compare GRoFA against five post-hoc debiasing methods applied to the same frozen backbone embeddings: LEACE, PASS/INLP, FairerCLIP, SFID, and the recent (2024–2025) Sparse-Autoencoder-based debiasing method debiaSAE [3] (per-backbone 4096-unit SAE trained on clean training embeddings, top-50 gender-correlated SAE features zero-clamped at inference). Accuracy values are severity-averaged (mean over levels 1–3 for each noise type). **Bold**: best, underline: second-best. **dS**: debiaSAE.

Table 1: CLIP ViT-B/16 on FairFace (α=0.80, 5-seed ensemble + WiSE-FT). BL: frozen CLIP baseline. Bold: best, underline: 2nd. 1st: first-place wins (out of 22 per task, 66 total). Stars on Ours: paired t-test Ours vs. Baseline ($^{*}p$<0.05, $^{**}p$<0.01).

Cond.	BL	LE	PA	FC	SF	dS	**Ours**
Race							
Clean	0.671	0.678	0.402	0.273	0.652	0.662	**0.682****
Gaussian	0.534	0.522	0.283	0.249	0.506	0.519	**0.584****
Shot	0.547	0.540	0.294	0.256	0.514	0.532	**0.586****
Impulse	0.473	0.456	0.224	0.215	0.444	0.428	**0.523****
Defocus	0.588	0.602	0.356	0.261	0.550	0.581	**0.654****
Glass	0.664	0.667	0.407	0.267	0.640	0.658	**0.677****
Cutout	0.573	0.599	0.305	0.258	0.559	0.605	**0.686****
Jpeg	0.633	0.652	0.371	0.271	0.607	0.640	**0.672****
1st	0	0	0	0	0	0	**22**
Gender							
Clean	0.952	0.952	0.949	**0.965**	0.952	0.944	0.952
Gaussian	0.837	0.836	0.827	**0.879**	0.829	0.820	0.874**
Shot	0.841	0.841	0.832	**0.881**	0.833	0.830	0.880**
Impulse	0.805	0.812	0.794	0.833	0.804	0.762	**0.842***
Defocus	0.916	0.916	0.914	**0.933**	0.914	0.859	0.920
Glass	0.944	0.944	0.941	**0.961**	0.943	0.935	0.944
Cutout	0.932	0.932	0.923	**0.957**	0.936	0.911	0.944
Jpeg	0.940	0.940	0.937	**0.949**	0.937	0.927	0.941
1st	0	0	0	**16**	0	0	6
Age							
Clean	0.530	0.526	0.522	0.479	0.515	**0.535**	0.518
Gaussian	0.439	0.439	0.433	0.430	0.433	0.423	**0.450***
Shot	0.434	0.434	0.426	0.423	0.421	0.436	**0.446****
Impulse	0.373	0.373	0.377	0.371	0.380	0.354	**0.401****
Defocus	0.456	0.457	0.452	0.451	0.459	**0.467**	0.455
Glass	0.528	0.529	0.523	0.471	0.522	**0.541**	0.520
Cutout	**0.487**	0.486	0.479	0.459	0.471	0.481	0.489
Jpeg	0.506	0.506	0.503	0.465	0.502	**0.508**	0.506
1st	0	1	0	1	0	**8**	12
Total 1st	0	1	0	17	0	8	**40**

A.2 WiSE-FT α Sweep

A.3 Fairness Metrics Comparison

B Statistical Significance Details

Table 10 reports Wilcoxon signed-rank tests and effect sizes for all backbone–dataset combinations, including per-task breakdowns.

Examining per-task patterns reveals a clear hierarchy of improvement consistency. **Race** classification shows the most robust improvement: all six backbone–dataset combinations achieve $p < 10^{-6}$ with large effect sizes ($d \geq 1.00$), confirming that the noise-gated adapter consistently strengthens race features under corruption. **Gender** is similarly significant across all settings ($p < 0.001$), though with slightly smaller absolute gains since the frozen-backbone baselines already achieve high accuracy on this binary task. **Age** exhibits backbone-dependent behavior: four of six settings are significant ($p < 0.01$), while two FairFace results—CLIP ($p = 0.091$) and DINOv2 ($p = 0.408$)—do not reach significance. These two cases are consistent with the known difficulty of FairFace's nine coarse age bins, where performance ceilings for all methods are low (42–53%), limiting the margin available for improvement.

C BLIP Per-Condition Detailed Results

Tables 11 and 12 report full per-condition accuracy (mean over 5 evaluation seeds) for the primary BLIP configuration on FairFace and UTKFace, respectively; Figure 2 of the main paper visualizes the same deltas as a heatmap. Each row corresponds to one of the 22 evaluation conditions: clean plus 7 noise types × 3 severity levels. **Bold** indicates the higher accuracy between Baseline and Ours; Δ denotes the signed improvement; p is the exact per-condition paired t-test p-value over the 5 evaluation seeds (the source of the two-tier significance markers in Tables 1 and 2 of the main paper).

D Training Dynamics

The GRoFA training procedure involves three adaptive mechanisms whose convergence behavior we describe here based on the recorded training histories.

Noise gate learning. The noise gate's predicted score starts near chance (≈0.49 at epoch 1) and converges to approximately 0.68 by epoch 25 (batch-average trajectory). The trajectory alone does not measure clean-vs-corrupted discrimination; per-condition separation is quantified in Table 13. The learned score then modulates the

Table 2: CLIP ViT-B/16 on UTKFace (α=0.60, 5-seed ensemble + WiSE-FT). BL: frozen CLIP baseline. Bold: best, underline: 2nd. 1st: first-place wins (out of 22 per task, 66 total). Stars on Ours: paired t-test Ours vs. Baseline ($^{*}p$<0.05, $^{**}p$<0.01).

Cond.	BL	LE	PA	FC	SF	dS	**Ours**
Race							
Clean	0.862	0.859	0.527	0.417	0.840	0.855	**0.869****
Gaussian	0.818	0.810	0.471	0.418	0.808	0.813	**0.856****
Shot	0.806	0.798	0.458	0.418	0.794	0.799	**0.861****
Impulse	0.799	0.804	0.454	0.419	0.786	0.804	**0.862****
Defocus	0.807	0.805	0.493	0.420	0.788	0.795	**0.851****
Glass	0.847	0.846	0.525	0.415	0.834	0.845	**0.866****
Cutout	0.846	0.847	0.504	0.416	0.828	0.832	**0.867****
Jpeg	0.855	0.854	0.522	0.418	0.841	0.849	**0.871****
1st	0	0	0	0	0	0	**22**
Gender							
Clean	**0.971**	0.970	**0.971**	0.965	0.970	0.944	0.969
Gaussian	0.963	0.963	0.966	0.960	0.964	0.930	**0.967***
Shot	0.964	0.964	0.966	0.958	0.964	0.930	**0.970***
Impulse	0.962	0.963	0.963	0.962	0.963	0.937	**0.968****
Defocus	0.934	0.939	0.934	0.955	0.938	0.938	**0.957***
Glass	0.966	0.966	**0.967**	0.964	0.966	0.944	0.967
Cutout	0.959	0.958	0.959	0.960	0.960	0.931	**0.967****
Jpeg	0.968	0.968	**0.969**	0.964	0.968	0.929	0.969
1st	1	0	5	1	1	0	**14**
Age							
Clean	0.596	0.596	0.588	0.424	0.593	0.598	**0.612***
Gaussian	0.506	0.508	0.499	0.428	0.517	0.503	**0.556****
Shot	0.522	0.521	0.512	0.425	0.529	0.521	**0.554****
Impulse	0.520	0.521	0.511	0.427	0.522	0.509	**0.565****
Defocus	0.508	0.510	0.503	0.407	0.510	0.494	**0.548****
Glass	0.583	0.582	0.573	0.422	0.581	0.578	**0.607****
Cutout	0.577	0.579	0.572	0.422	0.578	0.563	**0.600****
Jpeg	0.586	0.587	0.580	0.421	0.586	0.583	**0.602****
1st	0	0	0	0	0	0	**22**
Total 1st	1	0	5	1	1	0	**58**

Table 3: DINOv2-base on FairFace (α=0.70, 5-seed ensemble + WiSE-FT). BL: frozen DINOV2 baseline. Bold: best, underline: 2nd. 1st: first-place wins (out of 22 per task, 66 total). Stars on Ours: paired t-test Ours vs. Baseline ($^{*}p$<0.05, $^{**}p$<0.01).

Cond.	BL	LE	PA	FC	SF	dS	**Ours**
Race							
Clean	0.520	0.514	0.455	0.251	0.515	0.499	**0.554****
Gaussian	0.447	0.449	0.395	0.251	0.437	0.442	**0.507****
Shot	0.454	0.453	0.404	0.261	0.443	0.443	**0.512****
Impulse	0.359	0.375	0.302	0.224	0.349	0.340	**0.468****
Defocus	0.474	0.470	0.424	0.246	0.462	0.461	**0.542****
Glass	0.510	0.508	0.444	0.251	0.502	0.497	**0.547****
Cutout	0.477	0.481	0.427	0.243	0.483	0.476	**0.544****
Jpeg	0.514	0.507	0.448	0.251	0.507	0.485	**0.546****
1st	0	0	0	0	0	0	**22**
Gender							
Clean	0.897	0.897	0.898	0.897	0.898	0.878	**0.901***
Gaussian	0.859	0.859	0.860	**0.868**	0.860	0.848	0.861
Shot	0.862	0.863	0.865	**0.866**	0.860	0.850	0.864
Impulse	0.796	0.798	0.801	**0.840**	0.789	0.769	0.810**
Defocus	0.861	0.861	0.862	**0.873**	0.857	0.827	0.864*
Glass	0.890	0.890	0.889	**0.897**	0.890	0.872	0.890
Cutout	0.887	0.889	0.892	**0.901**	0.887	0.875	0.896*
Jpeg	0.893	0.893	0.893	0.895	0.892	0.879	**0.897****
1st	0	0	0	**18**	0	0	4
Age							
Clean	0.449	0.450	**0.451**	0.438	0.442	0.427	0.442
Gaussian	0.414	0.414	0.411	**0.422**	0.412	0.405	0.414
Shot	0.410	0.409	0.409	**0.421**	0.406	0.412	0.407
Impulse	0.392	0.393	0.393	**0.397**	0.390	0.369	0.392
Defocus	0.415	0.413	0.413	0.411	0.410	0.408	**0.415**
Glass	0.457	**0.458**	0.455	0.422	0.450	0.435	0.450
Cutout	0.417	0.416	0.427	**0.429**	0.419	0.424	0.424**
Jpeg	0.447	**0.450**	0.449	0.433	0.444	0.432	0.443
1st	2	5	5	**7**	1	1	1
Total 1st	2	5	5	25	1	1	**27**

Table 4: DINOv2-base on UTKFace (α=0.50, 5-seed ensemble + WiSE-FT). BL: frozen DINOV2 baseline. Bold: best, underline: 2nd. 1st: first-place wins (out of 22 per task, 66 total). Stars on Ours: paired t-test Ours vs. Baseline ($^{*}p$<0.05, $^{**}p$<0.01).

Cond.	BL	LE	PA	FC	SF	dS	**Ours**
Race							
Clean	0.756	0.742	0.622	0.433	0.749	0.723	**0.770****
Gaussian	0.727	0.719	0.593	0.435	0.721	0.694	**0.771****
Shot	0.721	0.714	0.584	0.434	0.718	0.690	**0.768****
Impulse	0.719	0.714	0.569	0.440	0.715	0.678	**0.771****
Defocus	0.707	0.694	0.583	0.422	0.702	0.670	**0.764****
Glass	0.746	0.734	0.611	0.433	0.742	0.711	**0.767****
Cutout	0.746	0.730	0.609	0.432	0.742	0.705	**0.774****
Jpeg	0.752	0.736	0.621	0.432	0.749	0.723	**0.770****
1st	0	0	0	0	0	0	**22**
Gender							
Clean	0.934	0.934	0.930	0.887	0.934	0.908	**0.943****
Gaussian	0.926	0.925	0.924	0.891	0.926	0.906	**0.942****
Shot	0.926	0.926	0.925	0.893	0.926	0.909	**0.944****
Impulse	0.921	0.920	0.919	0.881	0.920	0.900	**0.938****
Defocus	0.914	0.914	0.911	0.868	0.914	0.880	**0.935****
Glass	0.927	0.927	0.924	0.889	0.927	0.904	**0.941****
Cutout	0.933	0.932	0.924	0.885	0.933	0.905	**0.943****
Jpeg	0.934	0.933	0.931	0.890	0.933	0.906	**0.945****
1st	0	0	0	0	0	0	**22**
Age							
Clean	0.523	**0.524**	0.518	0.449	0.521	0.492	0.521
Gaussian	0.484	0.484	0.489	0.449	0.481	0.483	**0.498***
Shot	0.486	0.489	0.493	0.445	0.487	0.480	**0.500****
Impulse	0.497	0.498	0.500	0.446	0.494	0.476	**0.508****
Defocus	0.475	0.473	0.473	0.432	0.476	0.453	**0.489****
Glass	0.518	0.519	0.514	0.445	0.519	0.491	**0.520**
Cutout	0.518	**0.519**	0.515	0.450	0.518	0.481	0.516
Jpeg	0.524	**0.525**	0.518	0.449	0.524	0.495	0.524
1st	0	5	0	0	3	0	**14**
Total 1st	0	5	0	0	3	0	**58**

Table 5: WiSE-FT α sweep: CLIP ViT-B/16 on FairFace. BM>: baseline wins (/66). W: per-task wins (/22).

α	BM>	Race		Gender		Age	
	/66	W	Cln	W	Cln	W	Cln
0.50	51	22	68.1	15	95.0	14	52.3
0.60	51	22	68.3	16	95.0	13	52.1
0.70	53	22	68.2	17	95.1	14	52.0
0.80	54	22	68.2	18	95.2	14	51.8
0.90	53	22	68.3	18	95.2	13	51.8
0.95	53	22	68.3	18	95.2	13	51.7
1.00	54	22	68.3	18	95.2	14	51.7

Table 6: WiSE-FT α sweep: CLIP ViT-B/16 on UTKFace. BM>: baseline wins (/66). W: per-task wins (/22).

α	BM>	Race		Gender		Age	
	/66	W	Cln	W	Cln	W	Cln
0.50	60	22	86.8	16	96.8	22	61.0
0.60	62	22	86.9	18	96.9	22	61.2
0.70	62	22	87.0	18	96.9	22	61.6
0.80	62	22	87.0	18	96.9	22	61.9
0.90	62	22	87.0	18	96.9	22	62.0
0.95	62	22	87.0	18	96.9	22	61.9
1.00	62	22	86.9	18	96.9	22	61.9

Table 7: WiSE-FT α sweep: DINOv2-base on FairFace. BM>: baseline wins (/66). W: per-task wins (/22).

α	BM>	Race		Gender		Age	
	/66	W	Cln	W	Cln	W	Cln
0.50	49	22	55.4	17	90.0	10	44.4
0.60	49	22	55.6	18	90.1	9	44.3
0.70	50	22	55.4	19	90.1	9	44.2
0.80	50	22	55.6	18	90.1	10	44.0
0.90	47	22	55.5	17	90.0	8	43.9
0.95	47	22	55.5	17	90.0	8	43.8
1.00	46	22	55.4	16	90.0	8	43.9

Table 8: WiSE-FT α sweep: DINOv2-base on UTKFace. BM>: baseline wins (/66). W: per-task wins (/22).

α	BM>	Race		Gender		Age	
	/66	W	Cln	W	Cln	W	Cln
0.50	61	22	77.0	22	94.3	17	52.1
0.60	58	22	76.9	22	94.5	14	52.3
0.70	57	22	77.0	22	94.7	13	52.1
0.80	57	22	77.1	22	94.8	13	52.3
0.90	59	22	77.4	22	94.8	15	52.4
0.95	57	22	77.5	22	94.8	13	52.3
1.00	57	22	77.6	22	94.8	13	52.2

clean/noisy embedding mixture per input via Eq. (3) of the main paper.

Table 13 quantifies the converged gate's response on the evaluation splits: the Spearman correlation ρ between the gate value g (Eq. (3) of the main paper; stored for the seed-42 adapter before ensembling) and corruption severity, per corruption type, and the clean-vs-corrupted AUROC obtained when ranking images by g. When the clean level is included as severity 0, the gate value is positively rank-associated with severity for six of the seven corruption types; glass blur, a spatial corruption that preserves local pixel statistics, is the exception on FairFace. All values are recomputed from the stored evaluation embeddings by the released script `gate_analysis.py`.

RTDAR task weight evolution. The Routed Task-Decomposed Alignment Router begins with roughly equal task weights (approximately 0.42 / 0.14 / 0.44 for race / gender / age at initialization). Over the course of training, the weights evolve to approximately 0.37 / 0.20 / 0.43, reflecting the router's response to per-task loss plateaus: as race accuracy improves rapidly and its loss decreases, the router gradually shifts capacity toward gender, whose binary classification loss remains relatively higher due to the fairness regularization. Age retains the largest weight throughout training, consistent with its status as the most difficult multi-class task.

Fairness gap dynamics. The per-task accuracy gaps (max − min across demographic groups) show distinct trajectories. The race gap starts highest (≈0.05) and decreases steadily as the HSIC regularizer takes effect. The gender gap fluctuates more due to the interplay between classification loss and the stronger HSIC weight (λ_{race} = 0.5, λ_{gender} = 0.3). The age gap, which is not explicitly regularized via HSIC in our formulation, remains relatively stable throughout training.

E Hyperparameter Sensitivity

The key hyperparameters of GRoFA were selected based on held-out validation experiments on FairFace with the BLIP backbone. The CLS token ratio (0.7) determines the proportion of the adapter output that retains classification-relevant features versus noise-robust features; values in the range 0.6–0.8 produced comparable results, with 0.7 achieving the best balance. The HSIC regularization weights (λ_{race} = 0.5, λ_{gender} = 0.3) reflect the relative difficulty of achieving statistical independence between representations and each demographic attribute; race, with more categories, requires stronger regularization. The clean anchoring weight (0.5) prevents catastrophic forgetting of clean-image performance; this value was robust across the range 0.3–0.7, with extreme values (<0.2 or >0.8) causing either under-anchoring (clean accuracy degradation) or over-anchoring (insufficient noise robustness improvement). Importantly, these hyperparameters were fixed across all three backbones without per-backbone tuning, demonstrating the generality of the chosen configuration.

F Per-Backbone Training Details

To verify that GRoFA generalizes across backbone architectures, we trained identical adapter configurations on CLIP ViT-B/16 and DINOv2-base using the same hyperparameters as the primary BLIP configuration. The adapter architecture is a two-layer MLP (768 → 384 → 768) with GELU activation and layer normalization, applied identically to all three backbones. No per-backbone hyperparameter tuning was performed: the learning rate (1×10^{-4}), batch size (64), CLS ratio (0.7), HSIC weights (λ_{race} = 0.5, λ_{gender} = 0.3), and clean anchoring weight (0.5) were held constant (Table 14). The scalar weights were set by a small coordinate-style search on BLIP+FairFace only (learning rate over $\{1, 2, 3\}\times10^{-4}$, ArcFace weight over $[0, 1]$, CLS ratio over $[0.5, 1]$, HSIC weights over $[0.3, 0.5]$, anchoring over $[0.3, 0.7]$; see also Appendix E) and transplanted unchanged to UTKFace, CLIP, and DINOv2.

Training converged in 25–35 epochs across all configurations. Table 15 reports the per-seed accuracy ranges (before WiSE-FT ensembling), confirming low variance across random seeds.

The narrow accuracy ranges (within 1.2 percentage points across seeds for all configurations) confirm that GRoFA training is stable and does not require seed-specific tuning. The consistent convergence behavior across architecturally distinct backbones—BLIP (vision-language pretraining), CLIP (contrastive image-text), and DINOv2 (self-supervised vision)—supports the claim that GRoFA's adapter design is backbone-agnostic.

G Loss Leave-One-Out Ablation Details

To isolate the contribution of each loss family in the 17-loss GRoFA objective, we retrain five leave-one-out (LOO) variants with a single seed (s=42) on the BLIP backbone and evaluate without WiSE-FT or multi-seed ensembling (so every LOO row in Table 16 reflects the raw effect of a loss removal on the trained adapter). The "Full" row uses the same single-seed s=42, no-WiSE-FT protocol for a direct

Table 9: Clean-image fairness across three backbones. Gap: max−min per-group accuracy; WGA: worst-group accuracy. Negative ΔGap and positive ΔWGA indicate improvement.

Backbone	Dataset	Task	Acc_b	Acc_o	Gap_b	Gap_o	ΔGap	WGA_b	WGA_o	ΔWGA
BLIP+IPMix LoRA	FairFace	race	37.7	41.4	5.2	4.2	-1.0	35.2	39.4	+4.1
		gender	75.8	77.3	14.5	17.9	+3.3	68.5	68.8	+0.3
		age	36.6	31.6	12.3	12.0	-0.3	30.7	25.8	-4.9
BLIP+IPMix LoRA	UTKFace	race	77.8	77.1	0.9	1.0	+0.1	77.3	76.6	-0.8
		gender	91.8	92.2	10.1	7.6	-2.5	86.2	87.8	+1.6
		age	51.5	52.9	20.0	22.2	+2.2	45.9	45.4	-0.5
CLIP ViT-B/16	FairFace	race	67.1	68.2	1.4	0.9	-0.6	66.4	67.8	+1.4
		gender	95.2	95.2	4.6	5.8	+1.1	92.7	91.7	-1.0
		age	53.0	51.8	16.0	16.8	+0.8	44.8	44.4	-0.4
CLIP ViT-B/16	UTKFace	race	86.2	86.9	0.9	2.5	+1.6	85.7	85.7	-0.0
		gender	97.1	96.9	4.6	5.2	+0.6	93.8	93.3	-0.5
		age	59.6	61.2	19.3	16.8	-2.5	53.3	57.1	+3.7
DINOv2-base	FairFace	race	52.0	55.4	2.1	0.8	-1.4	51.0	55.0	+4.1
		gender	89.7	90.1	9.5	10.8	+1.2	85.6	84.7	-0.9
		age	44.9	44.2	9.7	11.9	+2.2	40.5	38.9	-1.7
DINOv2-base	UTKFace	race	75.6	77.0	1.0	0.6	-0.4	75.1	76.7	+1.6
		gender	93.4	94.3	5.6	5.8	+0.2	89.5	89.9	+0.4
		age	52.3	52.1	14.8	16.5	+1.7	48.2	47.5	-0.6

Table 10: Statistical significance across 6 backbone–dataset configurations, measured in the *single-seed+WiSE-FT* configuration. Wilcoxon signed-rank test on 66 paired conditions; 95% CI by paired bootstrap (10,000 resamples). The final 5-seed-ensemble+WiSE-FT configuration reported in the main paper is even more significant: p=8.6×10^{-11} **(FairFace) and** 7.4×10^{-12} **(UTKFace), one-sided.**

Backbone	Dataset	Scope	N	Wins	W	p	ΔAcc	Cohen's d
BLIP	FairFace	All	66	58/66	140	6.9×10^{-10}	+0.037	1.05 (large)
		Race	22	21/22	2	1.4×10^{-6}	+0.046	1.36 (large)
		Gender	22	19/22	9	1.4×10^{-4}	+0.032	1.04 (large)
		Age	22	18/22	36	0.002	+0.032	0.83 (large)
BLIP	UTKFace	All	66	58/66	118	2.9×10^{-10}	+0.029	0.72 (medium)
		Race	22	18/22	25	4.3×10^{-4}	+0.021	1.00 (large)
		Gender	22	18/22	31	0.002	+0.007	0.85 (large)
		Age	22	22/22	0	4.0×10^{-5}	+0.061	1.12 (large)
CLIP	FairFace	All	66	54/66	180	3.4×10^{-9}	+0.025	0.81 (large)
		Race	22	22/22	0	4.8×10^{-7}	+0.051	1.47 (large)
		Gender	22	18/22	18	4.3×10^{-4}	+0.018	0.89 (large)
		Age	22	14/22	74	0.091	+0.006	0.42 (small)
CLIP	UTKFace	All	66	62/66	18	3.6×10^{-12}	+0.025	1.21 (large)
		Race	22	22/22	0	4.8×10^{-7}	+0.035	1.71 (large)
		Gender	22	18/22	18	4.0×10^{-4}	+0.006	0.61 (medium)
		Age	22	22/22	0	4.8×10^{-7}	+0.032	2.18 (large)
DINOv2	FairFace	All	66	50/66	343	1.1×10^{-6}	+0.021	0.67 (medium)
		Race	22	22/22	0	4.8×10^{-7}	+0.060	2.33 (large)
		Gender	22	19/22	23	7.8×10^{-4}	+0.005	0.90 (large)
		Age	22	9/22	101	0.408	−0.001	−0.16 (n.s.)
DINOv2	UTKFace	All	66	61/66	41	1.0×10^{-11}	+0.020	1.15 (large)
		Race	22	22/22	0	4.8×10^{-7}	+0.037	2.06 (large)
		Gender	22	22/22	0	4.0×10^{-5}	+0.015	3.30 (large)
		Age	22	17/22	41	0.006	+0.007	0.78 (medium)
Aggregate		All	396	349/396	3723	6.1×10^{-55}	+0.028	0.89 (large)

Table 11: BLIP+IPMix LoRA on FairFace: full per-condition accuracy (5-seed ensemble + WiSE-FT α=0.95). Bold: higher of Baseline vs. Ours; p: exact paired t-test p-value.

		Race				Gender				Age			
Noise	Sev	BL	Ours	Δ	p	BL	Ours	Δ	p	BL	Ours	Δ	p
Clean	–	.377	**.414**	+.037	8.8×10^{-4}	.758	**.773**	+.014	0.04	**.366**	.316	−.050	2.1×10^{-4}
Gaussian	1	.279	**.351**	+.073	3.4×10^{-4}	.647	**.696**	+.049	1.1×10^{-3}	.275	**.298**	+.023	0.08
	2	.247	**.334**	+.086	3.8×10^{-5}	.624	**.669**	+.045	2.3×10^{-3}	.238	**.305**	+.067	2.7×10^{-3}
	3	.192	**.287**	+.095	1.1×10^{-4}	.601	**.660**	+.058	9.5×10^{-5}	.233	**.274**	+.041	1.3×10^{-3}
Shot	1	.290	**.359**	+.068	1.3×10^{-4}	.652	**.705**	+.052	1.5×10^{-3}	.283	**.306**	+.023	0.05
	2	.269	**.327**	+.058	7.8×10^{-5}	.626	**.685**	+.059	0.02	.251	**.291**	+.040	5.4×10^{-3}
	3	.198	**.305**	+.108	5.3×10^{-6}	.609	**.657**	+.047	4.9×10^{-3}	.244	**.298**	+.054	1.9×10^{-4}
Impulse	1	.235	**.351**	+.117	6.8×10^{-5}	.575	**.684**	+.109	9.3×10^{-5}	.200	**.296**	+.095	1.6×10^{-3}
	2	.214	**.322**	+.108	5.0×10^{-5}	.579	**.658**	+.079	1.8×10^{-4}	.177	**.271**	+.094	3.1×10^{-4}
	3	.182	**.305**	+.122	1.6×10^{-6}	.583	**.652**	+.068	5.4×10^{-4}	.178	**.274**	+.096	1.4×10^{-3}
Defocus	1	.347	**.352**	+.004	0.49	.702	**.710**	+.008	0.24	**.306**	.284	−.022	0.04
	2	.283	**.319**	+.036	3.5×10^{-4}	.643	**.674**	+.030	2.4×10^{-3}	.253	**.267**	+.014	0.05
	3	.268	**.295**	+.027	9.7×10^{-3}	.607	**.659**	+.052	6.7×10^{-4}	.237	**.274**	+.036	4.8×10^{-3}
Glass	1	.376	**.398**	+.022	0.05	.741	**.766**	+.024	3.5×10^{-3}	**.332**	.317	−.015	4.2×10^{-3}
	2	.344	**.363**	+.019	0.01	.697	**.725**	+.028	7.4×10^{-3}	.296	**.308**	+.012	0.08
	3	.325	**.326**	+.001	0.80	.663	**.687**	+.024	0.02	.273	**.275**	+.001	0.84
Cutout	1	.389	**.437**	+.048	1.7×10^{-4}	.745	**.795**	+.050	1.3×10^{-3}	.332	**.364**	+.032	4.7×10^{-4}
	2	.392	**.430**	+.039	1.8×10^{-4}	.750	**.786**	+.036	1.0×10^{-3}	.323	**.378**	+.056	2.0×10^{-4}
	3	.385	**.427**	+.042	3.3×10^{-3}	.751	**.776**	+.026	6.1×10^{-3}	.316	**.370**	+.054	1.1×10^{-3}
JPEG	1	.376	**.413**	+.037	1.2×10^{-3}	.758	**.773**	+.015	0.05	**.366**	.316	−.049	2.7×10^{-4}
	2	.339	**.358**	+.019	6.6×10^{-3}	.711	**.742**	+.031	3.6×10^{-4}	.252	**.286**	+.034	1.5×10^{-3}
	3	.280	**.296**	+.015	0.03	.673	**.687**	+.014	0.11	.189	**.270**	+.081	1.8×10^{-4}
Wins (/22)		**22/22**				**22/22**				**18/22**			

apples-to-apples comparison against each ablation; it therefore does not match the 62/66 headline figure in Tables 1 and 2 of the main paper, which aggregate 5 seeds and WiSE-FT interpolation.

The five LOO configurations set the following loss weights to zero while preserving all other hyperparameters: (1) *w/o Distillation*: disables the BLIP-KL distillation term (λ_{cls}=1.0, so all head-budget goes to CLS). (2) *w/o Classification*: disables per-task CE losses (λ_{cls}=0 and per-task weights = 0). (3) *w/o HSIC*: disables kernel independence penalties (λ_r=λ_g=0). (4) *w/o ArcFace+Adv*: disables angular margin and adversarial heads (ArcFace and adversary weights = 0). (5) *w/o Anchor+Noise*: disables clean anchoring and noise-level supervision (λ_{anc}=λ_{noise}=0).

Findings. (1) *Classification CE is indispensable*: removing the per-task CE heads collapses BM> by −29 on FairFace and −24 on UTKFace, confirming that the HSIC, ArcFace, and distillation terms cannot by themselves supervise the adapter to predict attributes. (2) *Clean anchoring + noise supervision together drive the noise gate*: removing both degrades BM> by −5 (FairFace) and −12 (UTKFace), with the largest drop on UTKFace gender (from 16 to 5) confirming that noise-level supervision is what teaches the gate to open/close in a task-appropriate way. (3) *ArcFace+Adversarial provides a modest but consistent lift*: disabling angular margin and adversarial heads costs −1 on FairFace and −6 on UTKFace; race accuracy survives on FairFace (22/22) but drops on UTKFace (from 17 to 15). (4) *HSIC has the smallest marginal BM> effect but still contributes to fairness*: removing HSIC costs 0 BM> on both datasets, yet the clean FairFace race accuracy gap widens from 4.2% (full model, Table 9) to 7.2% without HSIC, and the FairFace age gap tightens from 12.0% to 9.8%—indicating HSIC trades raw accuracy neutrality for per-group parity in a task-specific way. (5) *Distillation does not by itself improve BM> on a single seed* (FairFace +3, UTKFace +1): without KL distillation the adapter moves further from the frozen baseline and wins more conditions on raw single-seed eval. Distillation becomes important under the 5-seed ensemble + WiSE-FT setup, where it regularises the adapter into a subspace compatible with the frozen baseline—a prerequisite for WiSE-FT interpolation to yield coherent gains (see the ablation discussion in §4.5 of the main paper).

This ordering is consistent across both datasets: Classification is indispensable, Anchor and Noise together drive the gate, ArcFace and Adversarial provide a modest lift, and HSIC reshapes fairness without shifting raw accuracy. The multi-loss design therefore contributes complementary rather than redundant signal.

H Per-Loss FLOP Breakdown and Wall-Clock

This appendix expands the order-of-magnitude analysis in §3.6 of the main paper into concrete per-loss FLOP counts at the operating point used in our experiments (B=64, d=768, h_a=384, h_g=64, N_d=14,

Table 12: BLIP+IPMix LoRA on UTKFace: full per-condition accuracy (5-seed ensemble + WiSE-FT α=0.60). Bold: higher of Baseline vs. Ours; p: exact paired t-test p-value.

		Race				Gender				Age			
Noise	Sev	BL	Ours	Δ	p	BL	Ours	Δ	p	BL	Ours	Δ	p
Clean	–	**.778**	.771	−.007	0.12	.918	**.922**	+.004	0.08	.515	**.529**	+.014	0.01
Gaussian	1	.708	**.744**	+.036	$1.2{\times}10^{-4}$	.906	**.911**	+.005	0.18	.325	**.436**	+.111	$2.1{\times}10^{-4}$
	2	.686	**.721**	+.035	$2.1{\times}10^{-3}$	**.895**	.893	−.002	0.47	.281	**.414**	+.133	$5.4{\times}10^{-5}$
	3	.645	**.697**	+.052	$7.7{\times}10^{-4}$	.865	**.879**	+.015	0.01	.237	**.361**	+.124	$7.9{\times}10^{-6}$
Shot	1	.695	**.739**	+.044	$6.3{\times}10^{-5}$	.909	**.912**	+.003	0.37	.319	**.440**	+.121	$1.7{\times}10^{-4}$
	2	.663	**.711**	+.048	$5.7{\times}10^{-5}$	.888	**.896**	+.008	0.06	.258	**.393**	+.135	$8.7{\times}10^{-6}$
	3	.629	**.677**	+.048	$3.5{\times}10^{-5}$	.854	**.871**	+.016	$8.9{\times}10^{-3}$	.220	**.348**	+.128	$5.5{\times}10^{-5}$
Impulse	1	.704	**.728**	+.024	$6.6{\times}10^{-4}$	.887	**.906**	+.019	$2.9{\times}10^{-5}$	.359	**.449**	+.089	$2.8{\times}10^{-5}$
	2	.677	**.720**	+.043	$7.7{\times}10^{-4}$	.871	**.887**	+.016	$7.3{\times}10^{-3}$	.319	**.406**	+.086	$4.3{\times}10^{-5}$
	3	.643	**.700**	+.057	$4.3{\times}10^{-4}$	.851	**.874**	+.023	$9.9{\times}10^{-5}$	.269	**.382**	+.113	$8.5{\times}10^{-5}$
Defocus	1	.725	**.746**	+.021	$9.1{\times}10^{-4}$	.911	**.916**	+.005	0.22	.476	**.495**	+.019	$4.0{\times}10^{-3}$
	2	.665	**.704**	+.039	$7.5{\times}10^{-5}$	.855	**.876**	+.021	0.04	.386	**.404**	+.019	$1.8{\times}10^{-3}$
	3	.612	**.649**	+.037	0.01	.816	**.838**	+.022	0.02	.304	**.345**	+.041	$3.4{\times}10^{-3}$
Glass	1	**.766**	.764	−.002	0.48	.915	**.919**	+.004	0.30	.511	**.528**	+.017	0.02
	2	**.751**	.747	−.004	0.18	.904	**.913**	+.009	0.03	.487	**.515**	+.028	$5.9{\times}10^{-5}$
	3	.729	**.750**	+.021	$4.0{\times}10^{-3}$	.898	**.905**	+.008	0.09	.471	**.490**	+.019	0.01
Cutout	1	.770	**.773**	+.003	0.36	.920	**.929**	+.009	$6.1{\times}10^{-3}$	.511	**.539**	+.028	$1.7{\times}10^{-3}$
	2	.764	**.773**	+.009	0.03	.923	**.930**	+.007	$8.1{\times}10^{-3}$	.502	**.529**	+.027	$8.6{\times}10^{-3}$
	3	.769	**.770**	+.001	0.61	.912	**.922**	+.010	$1.9{\times}10^{-3}$	.502	**.523**	+.020	$6.8{\times}10^{-3}$
JPEG	1	**.779**	.771	−.007	0.12	.918	**.922**	+.004	0.13	.515	**.529**	+.014	0.02
	2	.762	**.769**	+.007	0.11	**.921**	.919	−.002	0.23	.502	**.517**	+.015	$9.5{\times}10^{-3}$
	3	.738	**.756**	+.018	0.01	.918	**.922**	+.004	0.03	.444	**.499**	+.055	$2.2{\times}10^{-4}$
Wins (/22)		**18/22**				**20/22**				**22/22**			

Table 13: Gate response on the evaluation splits: Spearman ρ between the gate value g and severity per corruption type, and clean-vs-corrupted AUROC.

	Gau.	Shot	Imp.	Def.	Glass	JPEG	Cut.	AUROC
FairFace	.29	.35	.50	.55	−.04	.36	.68	.73
UTKFace	.50	.49	.39	.46	.19	.37	.30	.75

S=5). We report theoretical FLOPs derived from the closed-form cost expressions; constants of 2 for multiply-accumulate are absorbed into the leading coefficient.

H.1 Per-Component FLOPs (Forward Pass)

H.2 Per-Loss FLOPs

The count column lists the individual scalar terms; the paper's "17" counts the 14 routed terms (rows 1–3) plus the CE, ArcFace, and HSIC families as three fixed-weight units, with the gating-tied BCE/anchor scalars and the domain adversary scoped outside the router. The Barlow Twins and VICReg redundancy terms are the largest components because they form $d{\times}d$ correlation matrices ($O(Bd^2)$); the B^2-scaled contrastive and HSIC terms follow, and everything else is linear or sub-linear in B and effectively negligible. Even the largest term remains $\sim10^4$ times smaller than one backbone forward pass. This is what justifies the design choice in §3.2 of the main paper: routing all 14 alignment terms through a single learned weighting (RTDAR) instead of pruning to a smaller subset costs essentially nothing per step, while it gives the router the freedom to up- or down-weight individual heads as training progresses.

H.3 Trainable Parameter Breakdown

H.4 Wall-Clock Estimates

Translating the FLOP counts above into wall-clock requires a concrete platform; we report estimates for a single NVIDIA A100 (312 TFLOPs/s peak in TF32, ~80% utilisation in mixed-precision adapter training of a frozen ViT). At B=64, one training step processes $2BC_{\text{enc}}+T_{\text{loss}} \approx 2.16{\times}10^{12}$ FLOPs (Table 17); at the assumed throughput this is roughly 10–12 ms/step. One epoch traverses all 88,000 pre-generated views (88,000/64 = 1,375 steps), i.e., ≈14–17 s/epoch under the A100 estimate. Measured wall-clock on our RTX 4090 (including data loading and the teacher forward) is ≈2.9 h per FairFace run and ≈2.0 h per UTKFace run at the 25–35-epoch convergence points reported in Table 15. The 5-seed ensemble is trained sequentially in our default pipeline but is embarrassingly parallel: S

Table 14: Full GRoFA hyperparameter table.

Parameter	UTK	FF
Architecture		
Adapter hidden dim	384	
Noise gate hidden dim	64	
Total trainable params	803K	805K
Trainable ratio	0.92%	
Loss weights		
CLS ratio (λ_{cls})	0.7	
HSIC race (λ_r)	0.5	
HSIC gender (λ_g)	0.3	
Clean anchor (λ_{anc})	0.5	
Noise sup. (λ_{noise})	0.1	
ArcFace weight	0.15	
Training		
Optimizer	Adam	
Learning rate	1×10^{-4}	
Router LR	10× base	
Batch size	64	
Max epochs	50	
Early stopping patience	20	
Num. races	5	7
Num. age groups	9	
Post-hoc calibration		
WiSE-FT α	0.50	0.95
Ensemble seeds	5	

Table 15: Per-backbone accuracy range across 5 seeds on clean test images (before WiSE-FT). All models use the same adapter architecture and hyperparameters.

Backbone	Dataset	Acc Range (%)	Epochs
BLIP	FairFace	49.3–50.2	30
BLIP	UTKFace	73.4–74.5	30
CLIP ViT-B/16	FairFace	68.6–69.2	30
CLIP ViT-B/16	UTKFace	80.7–81.1	25
DINOv2-base	FairFace	59.5–60.7	35
DINOv2-base	UTKFace	72.3–73.3	30

GPUs collapse the wall-clock back to one-seed time. Inference cost at B=64 is dominated by the S frozen-backbone forwards (∼10ms each), giving ≈50ms/batch for the 5-seed ensemble plus a <0.1ms WiSE-FT interpolation.

For deployment scenarios where the S=5 inference factor is unacceptable, the WiSE-FT-only configuration (single seed, α tuned on a held-out fold) recovers 54.4±1.9 baseline wins on UTKFace and 58.5±0.7 on FairFace (§3.3/§3.4 of the main paper), trading ∼4 baseline-wins for the 5× speedup.

H.5 Comparison with Baselines (Cost Summary)

The cost trade-off therefore separates into two axes. *(i) Training:* GRoFA’s per-step cost is asymptotically the same as full-encoder fine-tuning ($O(BC_{\text{enc}})$) but with 108× fewer trainable parameters;

Table 16: Loss leave-one-out ablation (BLIP, single-seed s=42, no WiSE-FT). BM>: #conditions beating the frozen BLIP baseline (/66). R/G/A: per-task BM> (/22). ΔBM>: change vs. the Full row.

FairFace	BM>	Acc	R	G	A	ΔBM>
Full (s=42, no WFT)	59/66	0.4489	21	20	18	—
w/o Distillation	62/66	0.4521	21	21	20	+3
w/o HSIC	59/66	0.4487	21	20	18	±0
w/o ArcFace+Adv	58/66	0.4533	22	20	16	−1
w/o Anchor+Noise	54/66	0.4481	20	18	16	−5
w/o Classification	30/66	0.4095	7	12	11	−29
UTKFace	BM>	Acc	R	G	A	ΔBM>
Full (s=42, no WFT)	51/66	0.6989	17	16	18	—
w/o Distillation	52/66	0.7009	17	14	21	+1
w/o HSIC	51/66	0.6986	17	16	18	±0
w/o ArcFace+Adv	45/66	0.6992	15	13	17	−6
w/o Anchor+Noise	39/66	0.6966	13	5	21	−12
w/o Classification	27/66	0.6745	9	9	9	−24

Table 17: Per-batch forward FLOPs (B=64, d=768, h_a=384, h_g=64). The frozen backbone is dominant; adapter, gate and heads add <0.01%.

Component	Closed form	FLOPs/batch
BLIP ViT-B/16 (frozen, fwd)	$B\,C_{\text{enc}}$	1.08×10^{12}
teacher copy (no grad)	$B\,C_{\text{enc}}$	1.08×10^{12}
Adapter (3 linear + GELU)	$2B(2dh_a + h_a^2)$	9.4×10^{7}
Noise gate (2 linear + sigmoid)	$2B(dh_g + h_g)$	6.3×10^{6}
Heads (race + gender + age)	$2Bd\sum_k K_k$	1.8×10^{6}
WiSE-FT interpolation	Bd	4.9×10^{4}
Total fwd / batch	—	$\approx 2.16\times10^{12}$
Adapter+gate+heads share	—	**< 0.01**%

Table 18: Per-batch loss FLOPs at the same operating point (B=64, d=768, K_r=7). The full objective adds ≈2×10^8 FLOPs/batch — more than four orders of magnitude below the backbone forward (Table 17).

Family	Closed form	Count	FLOPs/batch
Per-sample alignment (MSE, cos., SmoothL1, KL, FFL, L1, SSIM)	$O(Bd)$–$O(Bd\log d)$	7	$\approx 1.4\times10^{6}$
Batch contrastive/metric (NT-Xent, MultiSim, Triplet, ArcFace, Center)	$O(B^2d)$, $O(BK_rd)$	5	$\approx 2.0\times10^{7}$
Redundancy (Barlow Twins, VICReg)	$O(Bd^2)$	2	$\approx 1.5\times10^{8}$
HSIC (r, g)	$O(B^2d)$	2	1.3×10^{7}
Multi-task CE (post-head)	$O(B\sum_k K_k)$	3	$\sim10^{3}$
ArcFace logit adjust (fixed)	$O(BK_rd)$	1	$\approx 3.4\times10^{5}$
Domain adversary (GRL + focal)	$O(Bd)$	1	$\approx 2\times10^{5}$
Clean anchoring	$O(\lvert\mathcal{B}_c\rvert d)$	1	$\leq1.5\times10^{5}$
Noise BCE	$O(B)$	1	1.3×10^{2}
RTDAR router softmax	$O(N_d)$	1	$\sim10^{2}$
Total loss / batch	—	—	$\approx$ **1.9×10^{8}**
Loss share of fwd	—	—	$\approx$ **0.009**%

Table 19: Trainable parameter count per module. The 803K–805K total is 0.92% of the frozen BLIP encoder.

Module	Closed form	Params
Adapter W_1, W_2, W_3 + biases	$2dh_a + h_a^2 + (h_a{+}h_a{+}d)$	738,624
Noise gate W_4, W_5 + biases	$dh_g + h_g + h_g{+}1$	49,281
Heads (race / gender / age)	$d(K_r{+}K_g{+}K_a) + (K_r{+}K_g{+}K_a)$	≈ 14,000
RTDAR router $\mathbf{r}$	N_d=14 logits + 85 init params	99
Total (FairFace, K_r=7)	—	805K
Total (UTKFace, K_r=5)	—	803K
Frozen BLIP encoder	—	87M
Trainable ratio	—	0.92%

Table 20: Cost comparison vs. representative baselines. N: training-set size; B: batch size; S: ensemble seeds; C_{enc}: frozen backbone forward; Θ_{enc}: encoder parameter count.

Method	Train cost / step	Trainable params	Inference factor	Joint?
LEACE / INLP / PASS	closed-form ($O(d^2)$ once)	0	$1\times+O(d)$	No (fair.)
FairerCLIP	$O(N^2d)$ kernel fit	0	$1\times+O(d)$	No (fair.)
SFID	$O(Bd\sum_k K_k)$	$\Theta_{\text{tr}}\sim10^4$	1×	No (fair.)
AugMix / PixMix / IPMix	$O(BC_{\text{enc}})$+aug	Θ_{enc}=87M	1×	No (rob.)
GRoFA (ours)	$O(2BC_{\text{enc}}+N_dB^2d)$	Θ_{tr}=803K	$S\times$ (S=5)	**Yes**

the additional $O(N_dB^2d)$ term from the multi-loss objective is two orders below BC_{enc} at our settings. *(ii) Inference:* GRoFA pays $S\times$ a single forward in exchange for the variance-reduction provided by the embedding ensemble, matching the cost profile of a deep ensemble [2] or a model-soup deployment. Post-hoc projection methods are strictly cheaper at inference but do not address noise robustness; data augmentation is cheaper at inference but does not improve fairness. GRoFA is the only method in the comparison that pays *any* extra cost in exchange for the joint fairness+robustness objective, and that cost is bounded and explicit.

I Corruption Generation Parameters

Table 21 lists the exact per-severity parameters used to pre-generate the 21 corrupted views of every image (§3.7 of the main paper). The corruption family follows the ImageNet-C benchmark [1]; parameters were fixed once before all experiments and are identical for every method in the comparison. Pixel intensities are normalized to $[0, 1]$ before corruption and re-quantized afterwards.

Table 21: Per-severity corruption parameters (identical across datasets and methods).

Type	Parameter	Sev. 1	Sev. 2	Sev. 3
Gaussian noise	std. σ	0.08	0.12	0.18
Shot noise	photon scale c (Pois$(xc)/c$)	60	25	12
Impulse noise	salt-and-pepper fraction	0.03	0.06	0.09
Defocus blur	Gaussian-blur radius (px)	2	4	6
Glass blur	(σ, max shift, iters)	(0.7,1,1)	(0.9,2,1)	(1.5,2,2)
JPEG	quality factor	75	40	20
Cutout	square size (px)	30	50	70

J Training-Data Scaling

To probe whether the 4,000-image training protocol limits GRoFA, we scale the *adapter training pool* to 8,000, 16,000, and 32,000 FairFace images—sampled with the same builder and seed from the official train split *after excluding the* 1,000 *fixed test images* (85,744 candidates), so no test image enters any training pool—while holding everything else fixed: the frozen IPMix-LoRA backbone, all hyperparameters, the corruption parameters of Table 21 (22 views per image, all traversed every epoch), and the evaluation protocol (the same 1,000-image test set and clean-only probe training of §4.1 of the main paper). Each scale trains a single seed (s=42) and applies WiSE-FT at α=0.95, matching the single-seed configuration whose 4,000-image reference attains 58/66 baseline wins.

Table 22: Training-data scaling on FairFace (BLIP, single seed s=42 + WiSE-FT α=0.95, identical test set and hyperparameters). BM>: baseline wins out of 66; Acc: severity-averaged mean accuracy over the 66 conditions.

Training images	BM> (/66)	Mean Acc
4,000 (paper protocol)	58	0.449
8,000	51	0.437
16,000	59	0.445
32,000	62	0.455

The response is not monotonic: the 8,000-image pool dips below the 4,000-image reference before the trend recovers and improves through 16,000 and 32,000 images. The largest pool improves on the paper protocol by four baseline wins (58→62) at +0.006 mean accuracy, an eightfold data increase for a modest gain. Because each scale trains a single seed, part of the dip at 8,000 images may reflect seed-level variance rather than a systematic effect of pool size. We conclude that the 4,000-image protocol of the main paper is not the binding constraint on GRoFA: it already realizes most of the accuracy attainable with 8× more data, and the comparisons in the main paper measure the method rather than a shortage of training images.

References

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